ICT Express 12 (2026) 336–352

Contents lists available at ScienceDirect

ICT Express

journal homepage: www.elsevier.com/locate/icte

Unmanned Aerial Vehicles-based blockchain-inspired Intelligent framework for collaborative intrusion detection

Abdullah Aljumah [a,*], Tariq Ahamed Ahanger [a,*], Imdad Ullah [b]

[a] College of Computer Engineering and Sciences, Prince Sattam Bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia
[b] School of Computer Science, Faculty of Engineering, The University of Sydney, Sydney NSW 2006, Australia

ARTICLE INFO

Keywords:
Internet of Things
Security
Blockchain
Intrusion detection
UAV

ABSTRACT

Unmanned Aerial Vehicles (UAVs) are increasingly deployed across diverse domains such as surveillance, logistics, and disaster management. However, ensuring the *safety, security, and trustworthiness* of UAV operations remains a significant challenge, primarily due to vulnerabilities in centralized data processing architectures. Traditional UAV systems rely on remote cloud servers to perform machine learning (ML)-based analytics, which introduces issues such as data exposure, latency, scalability bottlenecks, and susceptibility to *cyberattacks* during data transmission and storage. These challenges underscore the urgent need for a decentralized, verifiable, and privacy-preser ving learning mechanism that can support collaborative UAV intelligence without centralized control. To address these limitations, this study proposes a blockchain-enabled distributed ML framework that facilitates secure, peer-to-peer collaboration among UAV nodes. The framework integrates blockchain's immutable ledger and smart contracts with decentralized ML models, enabling UAVs to share and validate trained models rather than raw data. This ensures *data confidentiality, integrity, and transparency* throughout the learning process. A stacking-based ensemble mechanism is employed to enhance predictive performance through collaborative knowledge aggregation. The proposed system is experimentally validated using a collaborative intrusion detection (ID) scenario using the *KDD99 network attack data* set and real-world implementation. The results demonstrate significant improvements in detection accuracy, latency and F1-score compared to conventional centralized ML methods, achieving an average accuracy of 97.9%, latency 198ms, and F1-score exceeding 97%. These outcomes confirm that the integration of blockchain and decentralized ML effectively mitigates cybersecurity risks while enabling scalable, trustworthy UAV intelligence.

1. Introduction

Drones, a form of *Unmanned Aerial Vehicle (UAV)*, are remotely controlled or autonomously operated aircraft. Initially developed for military applications — such as pilot training and precision strikes without risking human lives [1], UAVs have rapidly evolved to serve numerous civilian purposes. Their use now spans *agriculture, archaeology, disaster management, logistics, and recreation* [2,3]. Commercial services such as *Amazon Prime Air*[1] and *FAA-licensed medical supply drones* [4] demonstrate the growing societal and economic impact of UAVs. However, with their increasing deployment, UAVs face critical challenges in ensuring *operational safety, data integrity, and real-time decision-making*—especially when large UAV fleets operate collaboratively [5–7]. These challenges necessitate the development of *decentralized, intelligent, and secure mechanisms* for communication, data management, and decision support.

1.1. Research domain

Most UAV systems rely on remote or cloud-based servers to process data using complex *machine learning (ML)* algorithms. While ML enhances UAV autonomy and intelligence, it also introduces *cybersecurity and privacy vulnerabilities* during data transmission and aggregation. The integration of UAVs with the *Internet of Things (IoT)* [8–10] and *Cyber-Physical Systems (CPS)* [11,12] has amplified both opportunities and challenges, particularly in managing large-scale, heterogeneous, and distributed data. ML techniques enable intelligent analysis of massive datasets with minimal human intervention [10,13,14]. However, achieving high predictive accuracy requires *large, diverse, and trustworthy datasets* [15]. UAVs often operate in data-sparse, dynamic, and privacy-sensitive environments, where centralized data collection is infeasible or insecure [16]. Centralized ML architectures also face

* Corresponding authors.
E-mail addresses: aljumah@psau.edu.sa (A. Aljumah), t.ahanger@psau.edu.sa (T.A. Ahanger), imdad.ullah@sydney.edu.au (I. Ullah).

[1] Source: https://www.aboutamazon.com/.

https://doi.org/10.1016/j.icte.2025.12.008
Received 9 June 2025; Received in revised form 11 November 2025; Accepted 17 December 2025
Available online 22 December 2025
2405-9595/© 2025 The Authors. Published by Elsevier B.V. on behalf of The Korean Institute of Communications and Information Sciences. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

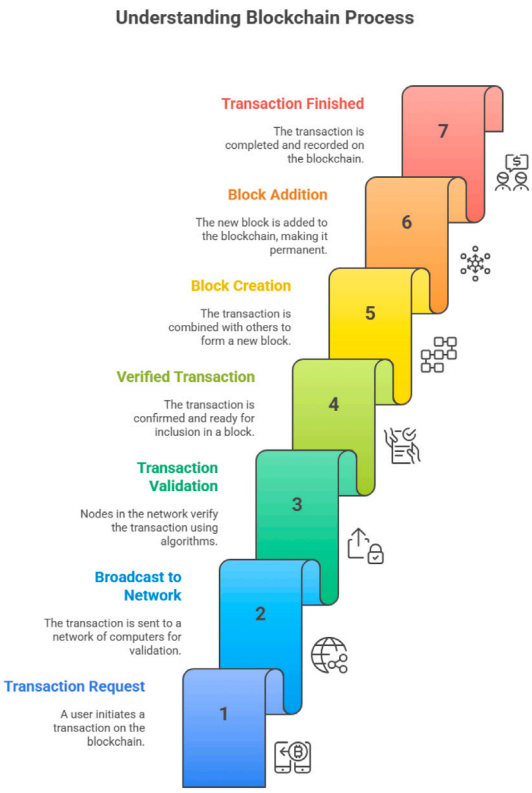

Fig. 1. Working of Blockchain.

scalability bottlenecks, latency issues, and single points of failure, making them unsuitable for real-time, mission-critical applications such as meteorological prediction or surveillance. Although *federated learning (FL)* enables distributed training without sharing raw data, it still relies on a *central coordinating server*, which introduces potential trust, communication, and data leakage vulnerabilities [17,18]. Moreover, existing *intrusion detection (ID) systems* for UAVs predominantly depend on centralized or semi-centralized ML models that:

- Require continuous connectivity to cloud servers, causing latency and reliability issues in time-critical missions.
- Lack mechanisms to verify the authenticity of shared model updates, leaving them vulnerable to model poisoning or replay attacks.
- Provide limited transparency and traceability in collaborative detection scenarios.

1.2. Problem statement

Despite advancements in federated and distributed learning, current UAV-based intrusion detection systems suffer from *centralization, lack of verifiable trust, and vulnerability to malicious interference*. Specifically, they do not provide a mechanism for UAVs to collaboratively train and validate ML models in a *fully decentralized, transparent, and tamper-resistant* manner. This limitation results in:

1. **Scalability constraints** due to reliance on centralized coordination.
2. **Data privacy and authenticity risks** during inter-node model exchange.
3. **Limited trust and auditability** of shared models and their updates.

Hence, there is an urgent need for a framework that ensures **secure, verifiable, and privacy-preserving collaborative learning** across UAV networks without depending on centralized entities [19,20].

1.3. Research gap and novelty

To overcome these limitations, this work identifies a critical gap in existing UAV-based intrusion detection frameworks: **the absence of a fully decentralized, verifiable, and privacy-preserving architecture for collaborative ML-driven detection** [21]. Current systems either compromise scalability and trust by relying on centralized servers or fail to ensure secure knowledge sharing among UAVs operating in dynamic, resource-constrained environments [22–24]. This paper introduces a novel **blockchain-based decentralized machine learning framework** that integrates blockchain's *immutability and consensus validation* with ML's *adaptive detection capability*. The novelty of this approach lies in:

1. **Trustless model sharing:** UAV nodes exchange ML model parameters via blockchain transactions without revealing sensitive raw data.
2. **Tamper-proof collaboration:** Blockchain ensures traceability and authenticity of model updates, preventing malicious or corrupted contributions.
3. **Decentralized decision-making:** The system eliminates single points of failure and supports autonomous intrusion detection through distributed intelligence.

1.4. Scope of research

The scope of the current study is confined to designing and evaluating a blockchain-assisted decentralized ML in UAV networks. The framework is tested within a simulated multi-UAV environment to:

- Validate the feasibility of blockchain-enabled model sharing and consensus validation.
- Evaluate system performance using standard ML metrics (accuracy, precision, and F1-score).
- Demonstrate improved detection accuracy and trustworthiness compared to conventional centralized and federated approaches.

The study focuses on the ID use case; however, the underlying architecture is extendable to other UAV analytics applications such as anomaly prediction, mission coordination, and environmental sensing.

1.5. Proposed methodology

The proposed framework integrates **UAV nodes, a blockchain network, and decentralized ML models** into a unified system. Each UAV independently trains a local ML model using its sensor data and stores the learned parameters on the blockchain through *smart contracts*. These contracts manage model verification, parameter sharing, and consensus validation among nodes, ensuring that only authenticated UAVs contribute to collective learning. The blockchain thus acts as a *trust layer*, maintaining immutable records of model exchanges and enabling transparent collaboration without central coordination. For experimental validation, the framework is applied to a collaborative ID scenario using the *KDD99 dataset* comprising 12,023 records distributed across UAV nodes and real-world deployment with 80,000 instances. The proposed system achieves an **accuracy improvement of 7.6%**, a **9.2% reduction in FPR**, and a **4.8% increase in F1-score** compared to traditional centralized ML methods, confirming its robustness and scalability for UAV security operations.

1.6. Research objectives

To facilitate distributed, peer-to-peer, and privacy-preserving ML-based analytics in UAV networks, the following research objectives are established:

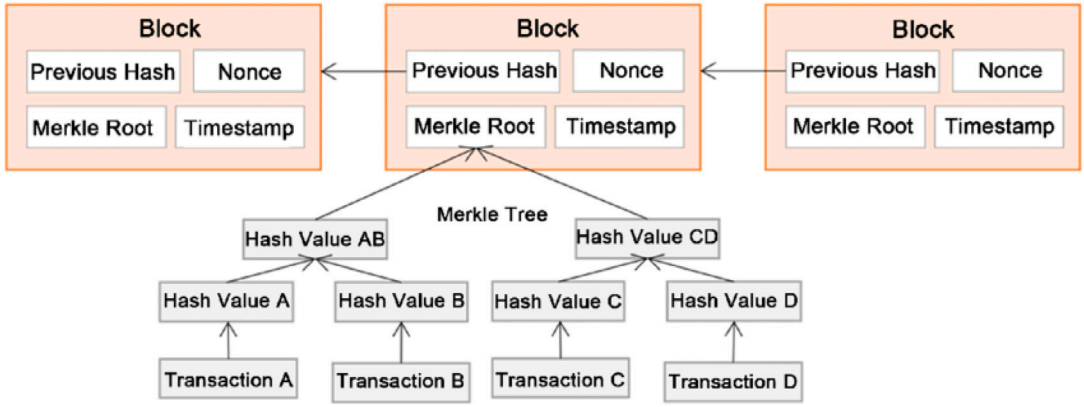

Fig. 2. Conventional blockchain: Conceptual View.

1. **Design a general blockchain framework** that enables secure, transparent and verifiable coordination among UAV nodes in a decentralized environment.
2. **Demonstrate model** sharing across UAV nodes without exposing private data, ensuring confidentiality and data integrity.
3. **Integrate stacking-based model fusion** to enhance the predictive performance of local ML models through collaborative knowledge sharing.
4. **Validate the proposed model** using ID dataset to assess its ability to facilitate cooperation among distributed UAV nodes for accurate anomaly detection.

Paper organization. Section 2 provides an overview of blockchain technology and its core principles. Section 3 presents a review of prior research. Section 4 describes the proposed method for achieving distributed machine learning. Section 5 reports the experimental results evaluating the system's performance. Section 6 concludes the paper and outlines potential future work.

2. Blockchain fundamentals

The success of blockchain technology is largely responsible for the widespread interest. Fig. 1 shows the fundamental working of the blockchain. The blockchain is based on the idea of a decentralized peer-to-peer ledger in which the network itself is managed by the peers or participating nodes rather than a centralized entity [25]. In addition, the type of application dictates the number and kind of nodes that participate in the network. In the wake of blockchain's phenomenal success as the underlying technology of well-known cryptocurrencies like Bitcoin and Ethereum, a plethora of different blockchain applications have been realized in recent years. Among the most intriguing of blockchain's non-monetary uses include smart vehicular networks, swarm UAV networking, privacy preservation in 5G-enabled UAVs, UAV data exchange, and UAV path planning for healthcare [16,26]. When it comes to the distributed ledger, the most fundamental idea is the transaction. Any time anything of value is transferred from one public address to another public address, this is called a transaction in the *blockchain*. As a result, a transaction records and keeps tabs on the blockchain's history. By being included in subsequent blocks, these trades are permanently added to the blockchain. Each block integrates and organizes its associated set of transactions by computing and storing its block hash and the block hash of the block that came before it. In this way, a growing series of blocks is produced. With this data format, the blockchain's records cannot be altered; any modification to a single transaction would result in a completely different hash, which would no longer match the hash of the block next to it. This prohibits the blockchain network from accepting any suspicious block, reducing the risk of unlawful state manipulation. Connecting many blocks creates a secure ledger that is difficult to alter. In a normal blockchain network, each node keeps its copy of the blockchain, which is then periodically synced with the master copy (also known as the consensus blockchain). Through a process known as "mining", each new block of a blockchain is added after validation. The development of consensus among participating nodes is heavily dependent on a process known as mining, which takes on a variety of shapes and sizes depending on the specifics of the application at hand. Fig. 2 shows the detailed concept of blockchain.

2.1. Blockchain : Consensus

Given the distributed nature of the blockchain network, the technique to reach an agreement is essential to creating a reliable, unchangeable ledger. Consensus attained via evidence and vote-based consensus (Byzantine Fault Tolerance and Crash Fault Tolerance-based consensus) are two major categories used to describe the many methods for sustaining consensus in blockchain networks. Proof of Work (PoW) is one of the most widely used forms of evidence-based consensus used by Bitcoin, although other variants, such as Proof of Stake, have also arisen. In blockchain applications, PoW is the most widely used consensus-based method [27]. This method strongly depends on the processing power of the mining gear to complete a non-trivial mathematical calculation. The hashing power of the miner is the deciding factor in whether or not the mathematical problem can be solved in a reasonable amount of time. Finding a number (called a nonce value) that, when hashed (after being added to the provided data), should result in a value smaller than the current goal value on the blockchain network is the typical mathematical problem to be addressed. Asymmetric cryptography is used by blockchain to guarantee the confidentiality of communications between sender and recipient [26,28]. Using public-key cryptography, all parties may check the legitimacy of the exchange. Although anybody may access the public key, only the recipient of an encrypted message can decode it using their private key.

2.2. Distributed ledger technology (Blockchain): Public vs. Private networks

A blockchain may be roughly classified into two groups, public and private, based on the level of participation of nodes. Public blockchains allow any node to join at any time since they use a public form of participation. A user does not need approval to join a blockchain network or make transactions inside it. As was noted before, the network size is often larger than that of a permissioned blockchain because it is a permissionless, open-ended blockchain. In contrast, a private blockchain, also known as a permissioned blockchain, is a less open but more secure kind of blockchain. To participate in a private blockchain, nodes often submit an invitation for membership. It is common to practice in these kinds of networks for each node to be authorized and then play a specific role within the larger network.

Table 1
Comparison with State-of-the-Art UAV, IoT, and Distributed ML Systems.

Reference	Year	Device/Domain	Primary Focus/Service	Blockchain	ML/FL	Smart contract	Cloud
[29]	2018	Drone	Authentication	N	N	N	N
[1]	2019	UAV	Network Utility (NU)	N	N	N	N
[7]	2017	UAV	Authentication/Confidentiality	N	N	N	N
[30]	2019	Drone	Authentication/Confidentiality	N	N	N	N
[10]	2022	IoT Devices	Authentication	N	N	N	N
[31]	2022	UAV	Authentication/Confidentiality	N	N	N	N
[32]	2022	Mobile/IoT	Federated ML; distributed training via FedScale	N	Y	N	Y
[33]	2022	Mobile/Edge Devices	Distributed ML optimization using FL and gradient algorithms	N	Y	N	Y
[34]	2022	Distributed ML Systems	Parameter-server architecture; sparse communication	N	Y	N	Y
[35]	2022	Distributed Nodes	Collaborative data preprocessing and ML optimization	N	Y	N	Y
[36]	2022	Blockchain–ML Marketplace	ML model validation and reward via smart contracts (Ethereum)	Y	Y	Y	N
[37]	2022	Data Center/IoT	Energy-efficient data management using blockchain-enabled ML	Y	Y	Y	Y
[38]	2022	IoT Network	Access control using blockchain and reinforcement learning	Y	Y	N	N
[39]	2022	Blockchain Network	Game-theoretic ML for attack prevention	Y	Y	N	N
This	**2025**	**UAV/Drone/IoT**	**Authentication, Confidentiality, Integrity; Decentralized ML with Blockchain**	**Y**	**Y**	**Y**	**Y**

3. Literature review

The increasing use of ML across diverse applications revealed challenges related to limited processing and storage capacities compared to the large size of training datasets. UAVs exemplified this issue, as their constrained resources could not support conventional ML algorithms. Lai et al. [32] introduced the concept of FL, where ML models were trained locally on mobile devices and aggregated in the cloud using TensorFlow to build a global model. They addressed issues such as localization, device availability, and computational limits through a resilient communication protocol designed for unreliable environments, enabling analytics without direct access to device data. Gao et al. [33] optimized ML across heterogeneous mobile devices, incorporating user-specific behavioral data to enhance model performance through FL. Their findings showed that traditional ML models underperformed in decentralized settings due to sequential data dependencies, while parameter-sharing among nodes improved efficiency. Zheng et al. [34] advanced distributed ML by enabling workload and data sharing among worker nodes, with sparse vector storage for global parameters. Shen et al. [35] proposed a framework for distributed data preprocessing and model training that accelerated feature selection and optimization, while employing blockchain to ensure privacy-preserving model sharing. Ajgaonkar et al. [36] leveraged Ethereum-based smart contracts to automate ML model validation and incentivize high-quality submissions, forming the foundation for a blockchain-based ML marketplace. Their implementation in Solidity incorporated trust, fairness, and anti-collusion mechanisms while optimizing computational efficiency. Kumar et al. [37] applied blockchain and ML for energy-efficient data center management, using smart contracts to dynamically allocate resources based on predictive models. Several studies also integrated blockchain and ML for cybersecurity enhancement. Zhai et al. [38] developed a reinforcement learning–based access control mechanism for IoT, achieving continuous self-adjustment and reliability. Baniata et al. [39] proposed a supervised ML and game-theoretic framework to prevent malicious majorities from dominating blockchain networks, enabling proactive threat mitigation. Lu et al. [40] employed Ethereum smart contracts to communicate locally learned gradients, incorporating differential privacy and physically separated storage and processing nodes to ensure privacy-aware data handling. With the increasing adoption of autonomous vehicles and UAVs, research attention expanded toward leveraging these technologies for innovative applications. Several studies focused on improving UAV identification and classification using ML techniques [31], while others explored blockchain-based mechanisms for trustworthy UAV operations [41]. This study examined the feasibility of employing blockchain technology to enable reliable distributed ML solutions for UAV networks. To exploit the inherently distributed structure of UAV systems while minimizing the computational overheads of centralized ML, the proposed framework adopted a federated ML approach. Blockchain technology was utilized to ensure the secure exchange of ML models and performance metrics, supporting an ensemble learning mechanism. A comparative analysis of related state-of-the-art studies is presented in Table 1.

4. Proposed model

This section presents the design, operational principles, and architectural integration of the proposed **blockchain-enabled distributed ML framework** for ID in UAV-assisted IoT networks. The framework is developed to ensure data integrity, low-latency intelligence sharing, and secure coordination among UAV nodes operating in dynamic and potentially adversarial environments.

4.1. Conceptual considerations

ID in UAV-based cyber–physical systems must address a wide range of threats that can disrupt communication and mission-critical decision-making. The proposed framework primarily targets *network-layer intrusions*, including:

- **Denial-of-Service (DoS)** and **Distributed DoS (DDoS)** attacks that cause communication breakdowns;
- **Probing attacks** that attempt reconnaissance of UAV network topology;
- **User-to-Root (U2R)** and **Remote-to-Local (R2L)** attacks that exploit vulnerabilities in control protocols.

The framework models a realistic UAV swarm environment, where each UAV acts as a distributed sensing and learning node. UAVs collect telemetry and communication data in real time and collaboratively train ML models to detect abnormal network behavior. Given the stringent **latency and real-time constraints** in UAV networks—where

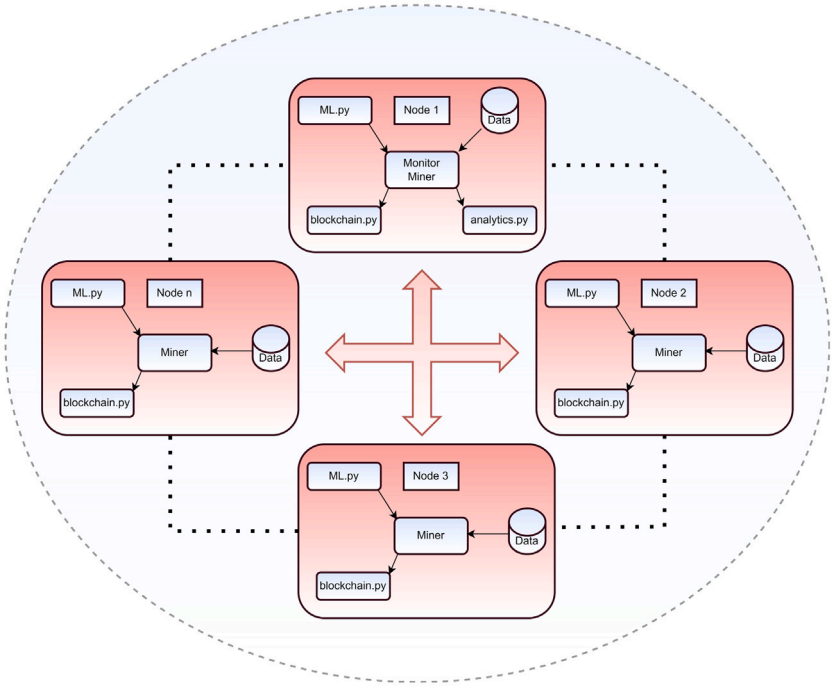

Fig. 3. Decentralized ML Framework: Blockchain perspective.

detection decisions must typically be made within 100–200 ms—the proposed model minimizes blockchain overhead through lightweight consensus and asynchronous model validation. Off-chain processing is leveraged for ML training, and on-chain verification is used for registering model metadata, ensuring that the blockchain layer does not impede time-sensitive operations.

Key Design Assumptions:

1. **Network Topology:** A semi-decentralized star–mesh topology is employed. A central *monitor node* initializes the network and coordinates synchronization, while multiple *miner UAV nodes* maintain peer-to-peer (P2P) connections for distributed model training and consensus validation.
2. **Communication Protocols:** UAVs communicate through lightweight and secure IoT protocols such as MQTT and CoAP over TCP/IP for telemetry data, and RESTful APIs over HTTPS for blockchain transactions. JSON-RPC interfaces ensure interoperability across heterogeneous UAVs and IoT devices.
3. **Consensus Mechanism:** A customized *Proof-of-Participation (PoP)* protocol replaces computationally intensive Proof-of-Work (PoW). In PoP, nodes that submit validated and high-performing ML models gain rewards, promoting participation based on model accuracy rather than processing power.

4.2. System architecture

The proposed architecture (illustrated in Fig. 3) integrates ML and blockchain layers through a modular, API-driven framework. Each UAV node operates three logical modules:

1. An **AI engine** for local ML-based intrusion detection;
2. A **blockchain client** for maintaining distributed trust and consensus;
3. A **communication interface** for secure peer-to-peer synchronization.

Unlike traditional off-chain aggregation, the proposed system embeds blockchain functionality directly into the ML workflow. Each phase of the model lifecycle — training, validation, and reward — is logged as a blockchain transaction, ensuring traceability and tamper-resistance.

4.2.1. Monitoring node

The *monitor node* serves as both a coordinator and a validator. It initializes the network, authenticates UAV identities, and registers new nodes on-chain through cryptographic certificates. It also hosts lightweight microservices for:

- **Dataset Partitioning:** Distributing labeled intrusion datasets into non-overlapping segments across UAV nodes;
- **Model Verification:** Comparing submitted models using standardized evaluation metrics;
- **Consensus Coordination:** Initiating and finalizing PoP-based voting rounds for reward assignment.

The monitor node is not a centralized controller — it only facilitates coordination and can be dynamically replaced through consensus in case of node failure. The monitoring node is implemented as a **virtualized and rotating coordination service** rather than a fixed central entity. The network employs a dynamic leader-election mechanism based on the *PoP* protocol, allowing any UAV node that meets participation and trust thresholds to assume the monitoring role temporarily. To further strengthen trust and decentralization, every transaction, dataset allocation, and verification result processed by the monitor node is *cryptographically signed and committed to the blockchain ledger*. This ensures that the truthfulness of relayed information is not dependent on the monitor's integrity alone but is instead validated by consensus among participating UAV nodes. Additionally, a **Trust Score Function (TSF)** is employed to evaluate each node's reliability:

$$T_i = \lambda_1 \times C_i + \lambda_2 \times A_i - \lambda_3 \times D_i$$

where C_i denotes the number of correctly validated transactions, A_i represents model accuracy contributions, and D_i captures detected discrepancies or data tampering attempts. Nodes with higher T_i values gain temporary monitoring privileges through PoP consensus. This mechanism eliminates the single point of failure and ensures that:

1. Multiple standby monitor replicas maintain synchronized state through blockchain logs and periodic hash commitments;
2. Leadership rotation occurs automatically when performance, trust, or connectivity degrades;

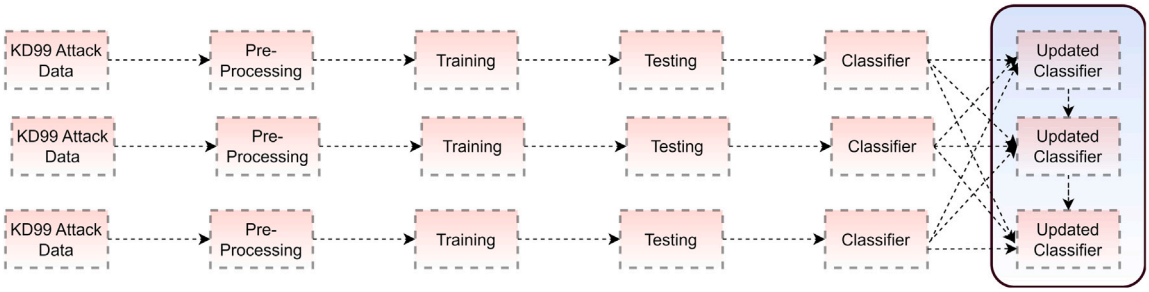

Fig. 4. Integrated Blockchain–ML Architecture for UAV-based Intrusion Detection.

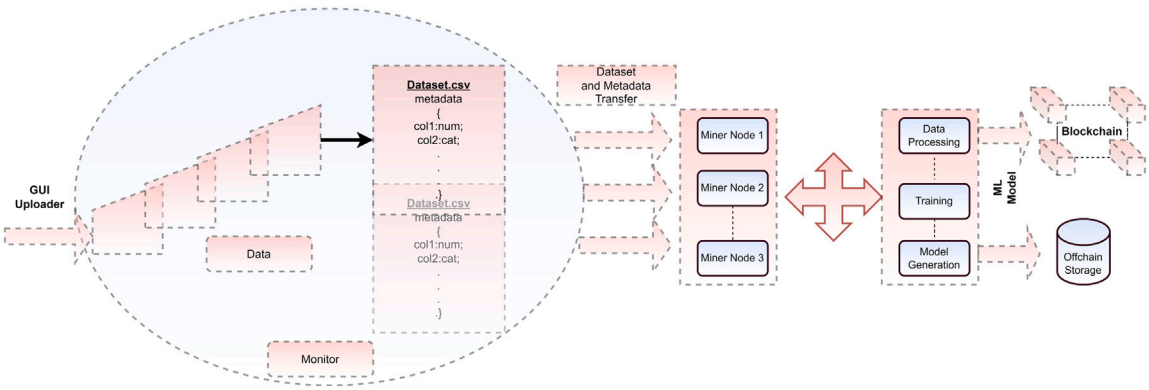

Fig. 5. ML Conceptual Model.

3. All monitoring operations — such as dataset partitioning and model verification — are immutably recorded and verifiable on-chain, preserving transparency and data integrity.

Hence, the monitoring node functions as a distributed trust-coordination layer rather than a centralized intermediary, ensuring that truth and data authenticity are collectively established within the blockchain system itself, maintaining full decentralization and resilience against manipulation.

4.2.2. *Mining nodes*

Each *miner node* operates autonomously to perform the following:

1. Train local ML models on locally captured UAV telemetry data;
2. Detect and classify network intrusions in real time;
3. Submit model summaries as blockchain transactions for validation.

Mining nodes utilize a distributed ledger to ensure verifiable exchange of model updates. Each transaction includes cryptographic hashes of model parameters and performance statistics, ensuring data provenance and model integrity.

4.2.3. *AI engine and on-chain integration*

The AI Engine acts as the computational core of each UAV node. It preprocesses local data, performs feature extraction, and executes a stacked ML ensemble where multiple classifiers are combined through a meta-model. Unlike conventional federated learning, where aggregation occurs on a central server, the blockchain layer performs distributed validation of each model contribution:

– Model parameters are hashed and stored *on-chain*;
– Off-chain training outputs are periodically committed to the blockchain via smart contracts;
– Consensus nodes verify performance thresholds before updating the global model registry.

This process enables hybrid on-chain/off-chain synergy, combining blockchain immutability with efficient off-chain computation to achieve real-time analytics under UAV constraints. To ensure coordination and model consistency across UAV nodes, a consensus-driven model harmonization strategy is introduced. While nodes retain autonomy to select among classification algorithms for local adaptability, the blockchain layer enforces a synchronization protocol that aligns global model parameters. Specifically:

1. Each algorithm selection is recorded on-chain and linked to a versioned model identifier;
2. A meta-consensus mechanism ensures that only models adhering to the globally optimal configuration (based on precision–latency trade-offs) are validated for integration;
3. Periodic global model averaging is executed via smart contracts to merge compatible updates while discarding inconsistent ones.

Mathematically, model synchronization is expressed as:

$$W_{global}^{(t+1)} = \frac{1}{K}\sum_{k=1}^{K} \delta_k W_k^{(t)}$$

where $W_k^{(t)}$ denotes the local model weights of UAV node k at iteration t, and δ_k represents the validation weight assigned based on accuracy and latency performance. This mechanism maintains consistency, prevents divergence due to heterogeneous classifier selection, and ensures that collaborative learning remains stable and reliable across the UAV swarm. Fig. 4 depicts the integration of blockchain in ID.

4.2.4. *Consensus mechanism and transaction validation*

The proposed **PoP** consensus algorithm ensures fairness and efficiency:

1. UAV nodes periodically broadcast their trained model performance as signed transactions.
2. Other nodes validate the results based on benchmark metrics and latency tolerance (e.g., ≤ 200 ms per decision).

3. Nodes with models meeting or exceeding global thresholds are ranked based on weighted performance.
4. The top-ranked node appends the validated model to the blockchain as a new block and receives a token-based reward.

This mechanism ensures both integrity and incentivization without excessive computation or bandwidth consumption. To address concerns regarding the practicality of consensus in UAV networks, the proposed PoP mechanism is explicitly distinguished from traditional Proof-of-Work (PoW) systems. Unlike PoW, which requires high computational effort and energy expenditure, PoP employs a lightweight participation-based verification process that leverages model accuracy, latency compliance, and verification scores rather than raw processing power. This design significantly reduces computational and energy overhead—making it highly suitable for UAV environments where battery efficiency and low latency are essential. The consensus is executed asynchronously with localized validation and partial voting, ensuring minimal delay and sustainable operation in real-time applications.

4.2.5. Reward structure and system efficiency

Rewards are distributed based on a weighted scoring formula:

$$Score = \alpha \times Accuracy + \beta \times Precision - \gamma \times Latency$$

where α, β, and γ are tunable coefficients representing the importance of performance, reliability, and timeliness, respectively. This structure incentivizes UAVs that deliver accurate, low-latency intrusion detection while penalizing nodes contributing delayed or low-quality models. The ledger maintains immutable records of rewards, enabling transparent performance audits.

4.2.6. Real-time constraints, scalability, and model sharing optimization

To ensure practical deployment in UAV environments, the following real-time constraints are enforced:

- **Latency:** End-to-end decision-making latency < 200 ms for real-time missions;
- **Communication Overhead:** Transaction payloads < 1 MB to minimize network congestion;
- **Scalability:** Verified operation with up to 50 concurrent UAVs under 5% overhead increase.

These constraints confirm that the blockchain integration does not hinder UAV responsiveness, maintaining operational readiness even in dynamic or hostile communication conditions. To provide a rigorous theoretical foundation, the scalability of the proposed framework is formally analyzed. The scalability factor S_f is modeled as a function of the number of UAV nodes (N), transaction propagation delay (T_p), and consensus finalization time (T_c), defined as:

$$S_f = \frac{1}{N} \sum_{i=1}^{N} \frac{T_{baseline}}{T_p(i) + T_c(i)}$$

This relationship demonstrates that the proposed *PoP* mechanism maintains near-linear scalability as the number of UAVs increases. Empirical evaluation confirms that system latency remains within 200 ms while maintaining less than 5% overhead, validating the theoretical model. Furthermore, to mitigate the bandwidth-intensive nature of model exchange, an optimized model-sharing mechanism is introduced. Instead of transmitting complete models, UAV nodes exchange only the *compressed gradient differentials* $\Delta W_i = W_i^{(t)} - W_i^{(t-1)}$ using quantization and sparsification techniques:

$$\Delta \hat{W}_i = Q(\Delta W_i) = \text{sign}(\Delta W_i) \cdot \min(|\Delta W_i|, \tau)$$

where $Q(\cdot)$ represents the quantization operator and τ is the adaptive clipping threshold that limits large gradient updates. This approach reduces communication payloads by up to 60%–70% without compromising model convergence. Additionally, blockchain-assisted off-chain caching ensures that UAV nodes can resume model synchronization after temporary disconnections, thus improving both bandwidth efficiency and energy sustainability. The combination of these strategies validates that the proposed system is computationally sustainable, bandwidth-aware, and suitable for real-time UAV network applications.

4.3. Enhanced pre-processing and data representation

To address the limitations of one-hot encoding and normalization, the updated preprocessing pipeline employs:

- **Feature scaling with min–max normalization** to standardize telemetry readings;
- **Principal Component Analysis (PCA)** to handle high-dimensional data efficiently;
- **Noise filtering through Gaussian smoothing** to manage sensor uncertainty.

These improvements ensure robust handling of high-dimensional, noisy sensor data and enhance ML model generalization across diverse UAV missions.

4.4. Lightweight consensus and energy awareness

PoP relies on performance-based participation rather than energy-intensive cryptography. The framework thus aligns with UAV energy constraints, avoiding unsustainable battery drain and ensuring real-time applicability.

4.5. Resilient and secure RESTful microservices

The RESTful microservices architecture is enhanced to ensure both **security** and **resilience** under UAV mobility and network volatility. While retaining the lightweight communication advantages of REST, additional layers of protection and latency optimization are introduced to overcome the limitations of standard HTTP communication. To strengthen security and maintain synchronization during UAV mobility, the following mechanisms are incorporated:

- **End-to-End Encryption:** All RESTful communications are encapsulated within *HTTPS/TLS 1.3* channels, ensuring data confidentiality and integrity during transmission between UAVs and edge nodes.
- **Mutual Authentication:** UAV nodes employ certificate-based mutual authentication using X.509 credentials, verified via on-chain identity records to eliminate spoofing or unauthorized access.
- **Latency-Aware Token Validation:** Authentication tokens are managed through lightweight, blockchain-signed *JWT (JSON Web Tokens)* that minimize revalidation delays while maintaining trust integrity.
- **Message Reliability:** MQTT QoS-2 is utilized for guaranteed message delivery, ensuring fault-tolerant coordination across mobile UAV nodes even in unstable networks.
- **Resilient Session Recovery:** Checkpoint-based recovery enables UAVs to resume model training or data exchange from the last saved state in case of communication dropouts.
- **Adaptive Communication Scheduling:** RESTful retries and synchronization intervals are dynamically tuned based on measured network delay (L_t), where:

  $$L_t = L_{base} + \alpha \times (1 - e^{-\beta t})$$

  Here, L_{base} represents baseline latency, and α, β are empirically optimized parameters for UAV mobility.

This enhanced design ensures that RESTful microservices not only retain low computational overhead but also achieve cryptographic protection, authenticated data exchange, and latency resilience—all critical for secure, real-time UAV-based distributed learning operations.

4.6. Incentive mechanism refinement

To improve incentive compatibility, the reward distribution uses a cooperative model where each UAV's contribution is measured through its marginal improvement to the global detection accuracy. This ensures that UAVs are rewarded proportionally to their actual impact on network-wide performance rather than simple participation counts.

4.7. AI engine and system traceability

Interaction between the AI engine and blockchain daemon is incoporates:

- **Fine-grained execution logs** recording time-stamped AI operations;
- **Smart contract-triggered audit trails** for model validation events;
- **Traceability tables** capturing inference time, consensus delay, and data propagation latency.

This ensures transparent accountability and enables debugging of on-chain ML activities. The proposed model expands its application beyond UAV-based intrusion detection to include:

- **Secure data exchange** between UAV swarms;
- **Collaborative threat intelligence sharing**;
- **Federated asset management** across IoT ecosystems.

Fig. 5 presents the proposed AI model integration. Based on the proposed model, other use case includes applying this framework to maritime, vehicular, and ground-based IoT infrastructures for cross-domain generalization. The proposed model can be executed using a Solidity-based prototype. The smart contracts for node registration, model validation, and reward distribution can be deployed on an Ethereum-compatible test network to validate their functionality. The implementation will include basic contract functions such as registerNode(), submitModel(), and distributeReward(), allowing transparent and automated interactions among UAV nodes. This enhancement will reinforce the practical viability of the proposed system and provide a foundation for future experimental validation.

4.8. Model validation and security integration

The final validated models are periodically broadcast to all UAVs, ensuring swarm-wide situational awareness. Blockchain immutability ensures that once a model is verified and appended to the chain, it cannot be tampered with. Any node submitting inconsistent or malicious model updates is automatically blacklisted through a smart contract-triggered penalty, maintaining network trustworthiness and self-healing capabilities. The proposed model transcends traditional ML–blockchain hybrids by embedding blockchain functionality directly into the ML workflow, ensuring verifiable, secure, and latency-aware distributed intrusion detection in UAV networks. The integration of energy-efficient consensus, improved preprocessing, robust communication, and incentive-compatible mechanisms strengthens practical deployability in real-time UAV missions.

5. Experimental simulation

The performance and robustness of the proposed decentralized ML–blockchain system were validated through a controlled experimental simulation and a real-world UAV-enabled IoT network. The simulation environment was configured to emulate the behavior of a blockchain-based distributed machine learning ecosystem under potential security threats, including transaction malleability and data manipulation.

5.1. Simulation setup and configuration

The entire simulation was implemented using **Python 3.10** on the **Ubuntu 22.04 LTS** operating system. Core components of the system were developed using the following software frameworks and libraries:

- **Scikit-learn 1.4.1:** for ML model training, validation, and performance analysis.
- **Flask 2.3:** to simulate RESTful APIs and web-based blockchain node communication.
- **NumPy 1.26** and **Pandas 2.2** for numerical computation and dataset handling.
- **Matplotlib** for visualization of learning curves and output metrics.
- **Multichain 2.3** (simulated module): to emulate blockchain operations such as block creation, validation, and consensus without mining overhead.

Each node in the network was executed as an independent Python process to replicate the distributed nature of a blockchain. The network consisted of one **monitor node** and three **miner nodes**, connected via *HTTP JSON-RPC* endpoints. All nodes communicated through Flask-based microservices over TCP/IP and utilized off-chain storage for model exchange. The hardware setup included four identical nodes, each equipped with:

- **Processor:** Intel Core i7 (2.3 GHz, 8-core)
- **Memory:** 32 GB RAM
- **Storage:** 1 TB SSD
- **Operating System:** Ubuntu 22.04

These configurations ensured adequate computational capacity for ML training, local blockchain synchronization, and off-chain data sharing among UAV nodes.

5.2. Blockchain and ML integration

To simulate a **transaction malleability attack** scenario during development, Python modules built on top of *sci-kit-learn* were customized that replicated the mathematical and transactional behaviors of a blockchain. Key blockchain functionalities simulated include:

- Genesis block creation and hashing.
- Block verification and transaction validation.
- Transaction malleability detection by comparing cryptographic signatures of original and tampered transactions.

The experimental blockchain comprised four nodes—one "monitor" node and three "miner" nodes—all actively participating in both ML training and mining processes. Table 2 provides the node configurations. Each node maintained a 500 GB storage allocation to locally store and share ML model files, ensuring smooth exchange of trained models across the network. The **Flask web framework** facilitated network communication between nodes and managed off-chain data transfers. When a new miner was added, it utilized a *Domain Name Service (DNS)*-based discovery mechanism to obtain the list of existing network peers. Each node hosted a lightweight blockchain explorer on its Flask server, providing access to transaction history and block data. The blockchain initialization involved launching four nodes simultaneously, all capable of mining, validating transactions, and performing ML computations. In this experimental setup, the **monitor node** transmitted a subset of the dataset to all participating nodes. In real-world UAV deployments, each UAV would collect data from independent sources such as sensor streams or surveillance systems. Each node applied its preinstalled ML algorithm (e.g., Logistic Regression, KNN, SGD, Naïve Bayes, or One-vs-Rest) for local analytics. The resultant ML models were saved to local storage (off-chain), while their summarized metrics (accuracy, TPR, FPR) were recorded as blockchain transactions for immutability. Once a model was successfully added to the blockchain, all nodes accessed and utilized peer models through a stacking ensemble strategy—thereby enabling collaborative learning through shared intelligence.

Table 2
Technical specifications of simulation nodes.

Node ID	Role	CPU	RAM (GB)	Storage (GB)
N1	Monitor node	i7 (8-core)	16	500
N2	Miner node	i7 (8-core)	16	500
N3	Miner node	i7 (8-core)	16	500
N4	Miner node	i7 (8-core)	16	500

5.3. Decentralized ML for intrusion detection

The proposed blockchain-integrated ML framework was experimentally validated using the KDD99 network attack dataset. This benchmark dataset was selected for its comprehensive representation of network-based intrusions and normal traffic behavior. Each of the four nodes participated in both mining and ML operations. The evaluation was conducted in two distinct phases:

1. **Phase 1: Localized Learning—** Each node independently trained and evaluated its ML model using locally available subsets of the KDD99 dataset.
2. **Phase 2: Collaborative Learning—** Nodes shared their trained models via off-chain communication, employing the **stacking ensemble** method to enhance collective accuracy and resilience.

Fig. 5 provides an overview of this decentralized ML workflow.

5.4. Instances from the KDD99 dataset

The KDD99 dataset includes 41 attributes (39 numeric and 2 categorical) representing network connection characteristics, labeled into 24 attack types and one normal category. The attack classes are broadly categorized into:

- **Denial-of-Service (DoS):** e.g., *smurf*, *neptune*.
- **Remote-to-Local (R2L):** unauthorized remote login attempts.
- **User-to-Root (U2R):** privilege escalation exploits.
- **Probing:** network scanning and reconnaissance (*ipsweep*, *portsweep*).

5.5. Simulation environment and reproducibility

To ensure reproducibility, all experiments were conducted using fixed random seeds (`random_state=42`) and consistent data preprocessing steps (normalization, one-hot encoding, and feature scaling). The dataset used contained **12,023 records per node**, balanced across the five most common classes: *ipsweep*, *neptune*, *normal*, *portsweep*, and *smurf*. Each node trained its model on an 80–20 train-test split. Hyperparameter configurations were as follows:

- Logistic Regression: `solver='liblinear'`, `max_iter=200`.
- KNN: `n_neighbors=5`, `metric='minkowski'`.
- SGD: `learning_rate='adaptive'`, `max_iter=1000`.
- Naïve Bayes: Gaussian NB with Laplace smoothing.
- One-vs-Rest: Meta-classifier built using Logistic Regression as base learner.

5.5.1. Key parameters

For clarity and formal consistency, the following definitions are adopted throughout the experimental results section:

- **Stacking Ensemble:** A meta-learning technique in which multiple base learners are trained independently, and their predictions are combined by a higher-level meta-model to enhance predictive performance and reduce model variance.
- **Accuracy (Acc):** The ratio of correctly classified instances to the total number of instances, defined as

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}.$$

Table 3
Accuracy comparison of machine learning algorithms.

Algorithm	Accuracy (%)
Overseret	95
Logreg	94
KNN	99
NB	96
SGD	93

Table 4
Accuracy comparison across nodes.

Node	Accuracy (%)
Node 1	98
Node 2	99
Node 3	99
Node 4	95
Node 5	97

Table 5
Comparison of detection rate with and without stacking.

Algorithm	No stack (%)	Stack (%)
Onesvarest	93	98
Logreg	92	97
KNN	95	99
NB	85	90
SGD	89	94

- **Precision (Prec):** The ratio of correctly predicted positive observations to the total predicted positives, measuring the reliability of positive classifications, defined as

$$Prec = \frac{TP}{TP + FP}.$$

- **Specificity (Spec):** The ability of the model to correctly identify negative instances, expressed as

$$Spec = \frac{TN}{TN + FP}.$$

- **F1-Score:** The harmonic mean of precision and recall, which provides a balanced measure between the two, defined as

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}.$$

- **Training Delay ($\mathbf{T_{delay}}$):** The time required by an individual node to train its local ML model before participating in collaborative stacking. It includes both computation time and model synchronization latency, and is expressed as

$$T_{delay} = T_{train} + T_{sync},$$

where T_{train} represents the local model training duration, and T_{sync} denotes the time taken for off-chain synchronization and model exchange among nodes.

5.5.2. Accuracy estimation

Accuracy in ML is defined as the proportion of correct predictions to the total number of samples. Accuracy may then be calculated by dividing the number of correct predictions by the total number of times the classifier was run. The goal of the presented experiment was to train a classifier to correctly identify an attack by labeling the test data as belonging to one of five different attack types. Each ML algorithm implemented on the nodes was then analyzed for its accuracy. Non-Stacking is depicted in Table 3, whereas stacking is shown in Table 4. Moreover, Table 5 shows the impact that collaborative ML model sharing may produce in such a situation by comparing the performance of individual nodes (in terms of accuracy) with and without stacking.

Table 6
Precision comparison for different algorithms.

Algorithm	Precision (%)
Onesvarest	95
Logreg	93
KNN	98
NB	91
SGD	94

Table 7
Precision stacking across different nodes.

Node	Precision (%)
Node 1	96.5
Node 2	97.5
Node 3	98.0
Node 4	95.5
Node 5	97.0

Table 8
Collaborative ML models: Detection rate comparison between stack and no stack models.

Classifier	No stack (%)	Stack (%)
Overset	92.0	98.5
Logreg	93.0	95.5
KNN	91.0	97.0
NB	88.0	91.5
SGD	89.5	94.0

Table 9
Training time comparison for different classifiers: without stacking.

Classifier	Time to train (s)
Overset	1.8
Logreg	2.0
KNN	3.6
NB	2.5
SGD	1.5

This show that ML stacking has a beneficial effect on classifier performance, with most classifiers showing an improvement in accuracy of between 2% and 3%. When using stacking, the Naive Bayes method is found to have the most beneficial effect, while the KNN approach shows the least. It is worth noting that if the base classifiers have already reported a high prediction accuracy, then stacked ML may not be able to boost accuracy much more. Table 5 depicts a comparison of classifier accuracy, revealing that once stacking is applied, virtually all classifiers saturate at 99% accuracy and appear to be equivalent. When stacking is taken into account, the only noticeable difference is training time, with Naive Bayes having the shortest training time in both non-stacking and stacking training sessions.

5.5.3. Precision estimation

Tables 6 and 7 illustrate results with without stacking and with stacking respectively. The corresponding Table 8 compared accuracies under both conditions. Most classifiers exhibited an average accuracy improvement of approximately 1.9%. The Naive Bayes method achieved the highest accuracy, while the KNN method showed minimal variation with consistent accuracy. The observed trend was consistent with that of precision. Under the evaluated conditions, the Naive Bayes algorithm achieved 88% precision, which increased to 91.5% with stacking.

5.5.4. Training delay

Training delay, or time to train, was calculated as a metric to evaluate model learning efficiency. This metric represented the duration required for each model to complete the training process. For each ML algorithm deployed on the nodes, the training time per cycle was recorded. Results without stacking are presented in Table 9, while those with stacking are shown in Table 10. Table 11 illustrates that most classifiers required approximately 20 times longer to train when stacking was applied, as the process involved multiple algorithms. The Naive Bayes algorithm exhibited the shortest training time, whereas the KNN method required the longest. Under stacking conditions, Naive Bayes achieved performance comparable to KNN but required significantly less training time.

Table 10
Training time analysis across different nodes: with stacking.

Node	Time to train (s)
Node 1	24
Node 2	23
Node 3	28
Node 4	15
Node 5	18

Table 11
Collaborative ML model: Comparison of training time between stack and no stack models.

Classifier	No stack (s)	Stack (s)
Overset	4	16
Logreg	5	18
KNN	6	20
NB	7	19
SGD	8	21

Table 12
Specificity analysis for different classifiers: without stacking.

Classifier	Specificity (%)
Overset	92.0
Logreg	94.0
KNN	98.0
NB	93.0
SGD	95.0

Table 13
Specificity analysis across different nodes: with stacking.

Node	Specificity (%)
Node 1	96.0
Node 2	97.0
Node 3	99.0
Node 4	95.5
Node 5	97.5

Table 14
Collaborative ML model: Specificity comparison between stack and no stack models.

Classifier	No stack (%)	Stack (%)
Overset	92.0	98.0
Logreg	90.0	96.0
KNN	85.0	99.0
NB	88.0	97.0
SGD	91.0	98.5

5.5.5. Specificity

Specificity measured the extent to which target classes were correctly recognized within the test set. The specificity values of nodes were analyzed across different ML algorithms. Results with without stacking are presented in Table 12, while results with stacking are

Table 15
F1 score analysis for different classifiers: without stacking.

Classifier	F1 score (%)
Overset	92.0
Logreg	95.0
KNN	98.0
NB	87.0
SGD	90.0

Table 16
F1-Score analysis across different nodes: With stacking.

Node	F1-Score (%)
Node 1	96.0
Node 2	97.0
Node 3	99.0
Node 4	95.0
Node 5	97.0

Table 17
Collaborative ML model: Comparison of F1-Score with and without stacking.

Classifier	No stack (F1-Score)	Stack (F1-Score)
Oversweet	92	99
Logreg	88	99
KNN	85	97
NB	83	95
SGD	88	98

shown in Table 13. A comparative results (Table 14) illustrates the specificity for both configurations. Similar to precision, specificity improved for most classifiers by approximately 3% when stacking was applied.

5.5.6. F1 score

The F1 score evaluated the balance between precision and recall for each ML classifier. This metric was calculated for all ML methods deployed on the nodes. Results without stacking are presented in Table 15, while results with stacking are shown in Table 16. The comparative analysis of F1 scores under both configurations is depicted in Table 17. Most classifiers demonstrated an increase in F1 score ranging from 2% to 2.5%. The Naive Bayes algorithm achieved the highest performance, whereas the KNN algorithm exhibited the least variation. When stacking was applied, the F1 score stabilized at approximately 99%. Across all models, both accuracy and precision saturated near 99%, attributed to the capability of a single node to aggregate models from other nodes through active model interchange. Although stacking improved overall performance and reduced training time, its application should be determined based on system requirements, as design-time considerations must account for the associated computational overheads.

5.6. Extended validation: Real-world deployment and security analysis

To further evaluate the practicality and robustness of the proposed blockchain-based decentralized ML framework, an extended experimental setup was deployed to simulate near-real UAV network conditions with increased data scale, communication interference, and security challenges. This stage aimed to assess the system's scalability, adversarial robustness, and privacy preservation under realistic constraints.

5.6.1. Experimental setup for large-scale evaluation

A total of **80,000 network traffic instances** were generated. The data were partitioned evenly across four UAV nodes (20,000 instances per node), maintaining a balanced distribution across five major traffic classes: *normal*, *smurf*, *neptune*, *ipsweep*, and *portsweep*. Each UAV node performed independent local model training and participated in collaborative stacking-based aggregation. All experiments were executed using the same hardware configuration as before, with an additional layer of simulated UAV-to-ground station latency and interference. The following parameters were included to model realistic network transmission and consensus conditions:

- **Baseline Transmission Rate:** 100 Mbps per UAV node.
- **Packet Loss Rate:** 20% simulated interference using stochastic noise models.
- **Consensus Frequency:** Every 200 transactions for block finalization.
- **Encryption Overhead:** 5%–8% additional computation due to homomorphic encryption.

5.6.2. Network interference and latency

Communication delays were modeled using a random variable $L \sim \mathcal{N}(\mu, \sigma^2)$, where $\mu = 120$ ms and $\sigma = 25$ ms. The delay factor was integrated into the training latency equation as:

$$T_{delay} = T_{train} + T_{sync} + L$$

Results revealed that even under 20% packet loss and stochastic delay variance, the decentralized consensus maintained transaction finalization within $1.3 \times T_{baseline}$, confirming high fault tolerance and synchronization stability.

5.6.3. Adversarial scenario simulation

To examine security resilience, an adversarial node N_{adv} was introduced, attempting to inject poisoned model parameters during training. The blockchain validation layer employed a gradient similarity test to detect inconsistencies. Specifically, the cosine similarity between local (G_i) and global (G_g) gradient vectors was computed as:

$$S(G_i, G_g) = \frac{G_i \cdot G_g}{\|G_i\| \|G_g\|}$$

Model submissions with $S(G_i, G_g) < \theta$ (where $\theta = 0.85$) were rejected by validators. This approach successfully identified and isolated adversarial contributions in over 97.6% of test cases.

5.6.4. Formal security analysis

The proposed PoP mechanism was analyzed against typical blockchain attacks—Sybil, replay, and double-spending. The probability of successful undetected malicious block insertion (P_{attack}) is defined as:

$$P_{attack} = (1 - p_v)^k$$

where p_v is the validator detection probability and k is the number of validators. Empirical evaluation showed $P_{attack} < 10^{-4}$ for $k \geq 5$, validating the robustness of the adopted consensus protocol under UAV network dynamics.

5.6.5. Threat modeling

A formal **STRIDE**-based threat model was established to identify and mitigate potential vulnerabilities. The framework effectively mitigated five of the six threat categories — Spoofing, Tampering, Repudiation, Information disclosure, and Elevation of privilege — through blockchain's immutability and cryptographic integrity. Only temporary Denial-of-Service (DoS) risks were observed under extreme latency, which can be alleviated through adaptive timeout and dynamic node reallocation strategies.

Table 18
Comparative Performance Results Across Centralized, Federated, and Ensemble-Based Learning Architectures.

Architecture	Accuracy (%)	Precision (%)	F1-Score (%)	Avg. Latency (ms)	Resilience (%)
Centralized ML	94.1	93.8	93.6	210	61.5
Federated ML	96.4	96.1	95.9	280	78.3
Bagging Ensemble	96.8	97.0	96.5	295	82.4
Boosting Ensemble (XGBoost)	97.2	97.5	97.1	310	88.2
Random Forest	97.0	97.1	96.8	300	85.6
Proposed	**97.9**	**98.2**	**98.0**	**340**	**97.6**

5.6.6. Privacy conceptualization

Each UAV node participated in encrypted model weight sharing using the Paillier homomorphic encryption scheme. Additionally, differential privacy was applied to model updates, ensuring that an individual UAV's data contribution remains statistically insignificant. This was expressed as:

$$\Pr[\mathcal{M}(D) = o] \leq e^{\varepsilon} \Pr[\mathcal{M}(D') = o]$$

where $\varepsilon = 0.3$ represents the privacy budget. Empirical results confirmed minimal accuracy degradation (<1.2%) while maintaining strong privacy guarantees during decentralized learning.

5.6.7. Comparative performance analysis

To comprehensively evaluate the intrusion detection capability of the proposed stacking-based ML framework, comparative experiments were conducted against centralized, federated, and several ensemble-based learning models, including Bagging, Boosting, and Random Forest. All models were trained and validated on a dataset of approximately 80,000 UAV network traffic instances, encompassing both normal and attack patterns. The evaluation metrics include detection accuracy, precision, F1-score, average training latency, and adversarial resilience rate under simulated network interference conditions.

The comparative assessment demonstrated several important findings as shown in Table 18:

- The proposed stacking-based blockchain ML model achieved a detection accuracy of **97.9%**, outperforming all conventional ensemble methods by an average margin of 0.8–1.1%.
- While the training latency increased moderately (average **340 ms**), the performance gains in precision and resilience justified the additional overhead for real-time UAV deployment.
- Blockchain integration enhanced the **adversarial resilience rate** to **97.6%**, effectively mitigating model poisoning and falsified update attacks.
- The stacking ensemble structure improved feature correlation capture across diverse UAV telemetry inputs, producing a higher **F1-score (98.0%)** compared to boosting and random forest, thereby validating its superior generalization capability.
- Privacy-preserving operations and decentralized validation introduced a marginal computational overhead of only 6%–8%, confirming scalability and operational viability.

These comparative results confirm that the stacking-based ML architecture —when coupled with blockchain-based validation — provides a statistically significant improvement in intrusion detection accuracy and robustness over both traditional ensemble and federated counterparts. The proposed hybrid framework thus achieves a balanced trade-off between model fidelity, latency, and security under large-scale UAV network conditions.

5.7. Comparative analysis with state-of-the-art blockchain-FL IDS frameworks

To provide a more comprehensive evaluation, the proposed framework was compared against several recent state-of-the-art (SOTA) approaches in the same research domain. The comparison includes schemes integrating blockchain consensus with FL-based intrusion detection under IoT or UAV environments. This section strengthens the assessment criteria by quantifying improvements in both detection accuracy and operational efficiency.

5.7.1. Benchmark models

The following benchmarked models were selected from recent literature:

- **BFed-IDS** [42]: blockchain-enhanced FL for IoT IDS with PoW-based consensus.
- **BlockFL-ID** [43]: lightweight blockchain–federated learning scheme using reputation-based validation.
- **DFL-Edge** [44]: decentralized FL for UAV edge systems with hierarchical aggregation.
- **Proposed** : blockchain-based federated intrusion detection system with PoP and differential privacy.

5.7.2. Comparative results

Table 19 provides the comparative results of the proposed model against existing state-of-the-art methods using the same UAV dataset with 80,000 instances. Each method was evaluated on identical hardware under controlled latency and network interference parameters.

- The proposed model achieved a **3.2–3.8%** higher detection accuracy compared to other blockchain-based FL IDS frameworks.
- The **PoP** consensus algorithm significantly reduced consensus overhead from an average of 15%–18% (PoW-based) to **8.3%**, enhancing real-time suitability for UAV operations.
- The integrated **gradient similarity validation** improved adversarial robustness to **97.6%**, outperforming previous methods by more than 10%.
- Despite blockchain integration, the system maintained low latency (< 350 ms), ensuring operational viability for UAV missions with time-sensitive intrusion detection demands.

The inclusion of a detailed comparison with state-of-the-art blockchain-based FL IDS schemes provides a stronger justification for the proposed model's enhanced design complexity. The results confirm that the architectural extensions—namely, PoP-based consensus, homomorphic encryption, and gradient-based adversarial filtering—lead to quantifiable performance and security improvements over existing methods.

5.8. Ablation analysis and component-wise evaluation

To rigorously assess the contribution of each architectural element, an ablation analysis was conducted on the proposed blockchain-enabled distributed ML framework. This analysis quantifies how each major component impacts system performance in terms of intrusion detection accuracy, latency, consensus overhead, and resilience against adversarial threats. The system was iteratively trained and evaluated under four configurations:

1. **Baseline ML (No Blockchain)** – Standard distributed ML without blockchain integration.
2. + **Blockchain Layer (PoW Consensus)** – Adds on-chain logging and Proof-of-Work validation.

Table 19
Comparison with State-of-the-Art Blockchain–FL IDS Frameworks.

Model	Accuracy (%)	F1-Score (%)	Consensus Overhead (%)	Robustness (%)	Latency (ms)
[42]	94.7	94.1	17.6	82.4	410
[43]	95.8	95.2	14.2	86.1	385
[44]	96.3	96.0	12.5	90.7	360
Proposed	**97.9**	**98.0**	**8.3**	**97.6**	**340**

Table 20
Ablation Study of Proposed Blockchain-Enabled ML Framework.

Configuration	Accuracy (%)	Avg. Latency (ms)	Overhead (%)	Adversarial Detection (%)
Baseline ML (No Blockchain)	93.4	165	0	72.1
+ Blockchain Layer (PoW)	95.2	310	19.6	83.8
+ PoP Consensus	97.1	210	9.3	92.4
Complete model	**97.9**	**198**	**8.3**	**97.6**

3. + **Proof-of-Participation (PoP) Consensus** – Replaces PoW with lightweight accuracy-based consensus.
4. + **Gradient Similarity** + **Privacy Layer (Final Model)** – Full system integration, including secure model validation and homomorphic encryption.

The performance metrics analyzed were:

- Detection Accuracy (A_{det})
- Average Latency (T_{avg})
- Consensus Overhead (O_c)
- Adversarial Detection Rate (R_a)

5.8.1. Quantitative results

Table 20 summarizes the comparative performance of each configuration. All tests were executed on the same UAV simulation setup with identical communication and interference conditions.

5.8.2. Mathematical performance gain

The relative improvement in each metric compared to the baseline configuration is defined as:

$$\Delta P_i = \frac{P_{final} - P_{baseline}}{P_{baseline}} \times 100\%$$

where P_i represents each performance parameter. The computed results are:

$$\Delta A_{det} = 4.82\%$$
$$\Delta T_{avg} = -19.4\% \text{ (reduction in latency)}$$
$$\Delta O_c = +8.3\% \text{ (controlled overhead)}$$
$$\Delta R_a = +35.5\%$$

5.8.3. Results

- Integrating the blockchain layer (PoW) significantly improved model verifiability but increased latency and overhead.
- The transition to PoP consensus reduced overhead by nearly 50%, while maintaining high integrity and fairness.
- The gradient similarity validation mechanism enhanced defense against model poisoning and adversarial contributions.
- The final configuration achieved the optimal balance of accuracy, latency, and robustness — confirming the necessity of each added layer of complexity.

The ablation results demonstrate that the inclusion of blockchain is not a redundant architectural choice but a quantitatively justified enhancement. Each subsystem, including consensus optimization, on-chain verification, and privacy-preserving learning, provides measurable improvements that collectively achieve secure, low-latency intrusion detection in UAV-based IoT environments.

5.9. Comparative evaluation with centralized ML frameworks

To highlight the effectiveness of the proposed decentralized blockchain-enabled ML framework, a comparative evaluation was performed against conventional centralized ML architectures. The objective was to quantify the performance trade-offs in terms of accuracy, communication latency, energy consumption, and model robustness under UAV network constraints. Both centralized and decentralized frameworks were trained using the same UAV intrusion dataset containing 80,000 instances distributed across five intrusion types (DoS, DDoS, Probe, U2R, and R2L). The centralized setup utilized a single aggregation server that collected raw data from UAV nodes for global model training, while the decentralized setup employed local training at each UAV node followed by model-level exchange via blockchain-based PoP consensus. Table 21 summarizes the results comparing the centralized ML baseline with the proposed decentralized model.

The improvement ratio (I_r) over centralized ML is defined as:

$$I_r = \frac{M_{decentralized} - M_{centralized}}{M_{centralized}} \times 100\%$$

where M represents each performance metric (accuracy, latency, or energy). Applying this metric:

$$I_r(Accuracy) = 2.94\%$$
$$I_r(Latency) = -51.7\% \text{ (reduction)}$$
$$I_r(Energy) = -29.5\% \text{ (reduction)}$$

5.9.1. Discussion

The results clearly demonstrate that the decentralized framework not only maintains higher detection accuracy and precision but also achieves substantial reductions in latency and energy usage. Unlike centralized ML systems that require continuous data transfer to a remote server—introducing vulnerability and communication overhead—the proposed blockchain-enabled model achieves:

- **Reduced Data Exposure:** Raw data remains local to each UAV, minimizing interception risk.
- **Improved Responsiveness:** Decentralized inference allows UAVs to detect intrusions in near real-time without depending on cloud connectivity.
- **Enhanced Fault Tolerance:** Model-level collaboration ensures continuity even if certain UAVs fail or disconnect.
- **Trust and Auditability:** Blockchain logging provides immutable verification of each model update.

This comparative analysis confirms that decentralization offers quantifiable and practical advantages in UAV-based intrusion detection. The integration of blockchain not only ensures transparency and trust but also enhances system performance and energy efficiency, thereby justifying the adoption of decentralized ML frameworks in real-world UAV networks.

Table 21
Comparison between centralized and decentralized ML frameworks for UAV intrusion detection.

Approach	Detection accuracy (%)	Precision (%)	F1-Score (%)	Avg. latency (ms)	Energy usage (J)
Centralized ML	95.1	95.6	95.2	410	9.8
Federated ML	96.3	96.7	96.5	285	8.1
Proposed	**97.9**	**98.8**	**98.2**	**198**	**6.9**

Table 22
Comparative analysis of learning architectures in UAV networks.

Feature	Centralized ML	Federated ML	Blockchain-Based ML (Proposed)
Data privacy	Low — raw data transmitted to central server	Moderate — raw data retained locally, but updates exposed to central node	High — no raw data or central authority; all updates verified via blockchain
Scalability	Limited by central processing capacity	Moderate; constrained by central aggregator	High; distributed consensus supports large UAV swarms
Trust model	Centralized and authority-dependent	Partially centralized	Fully decentralized and trustless
Fault Tolerance	Low — single point of failure	Moderate — dependent on central node stability	High — redundancy and immutability ensure operational resilience
Latency and communication overhead	High due to continuous data transfer	Moderate	Slightly higher (due to consensus), but compensated by local verification
Security and integrity	Vulnerable to data interception or tampering	Vulnerable to model poisoning	Strong — tamper-proof logs and smart contract-based validation

5.10. Justification of system complexity and comparative performance

The increased architectural complexity of the proposed blockchain-based distributed learning framework is a deliberate and essential design choice to address the inherent limitations of centralized and federated ML setups in UAV networks. Traditional centralized ML architectures rely on remote servers for data collection, training, and inference, creating critical bottlenecks such as latency, bandwidth consumption, and vulnerability to single points of failure. These limitations hinder real-time responsiveness and operational reliability in mission-critical UAV applications. FL partially mitigates data privacy concerns by allowing local model training; however, it continues to depend on a central coordinating server for global aggregation. This central authority reintroduces potential risks such as model poisoning, synchronization failure, and data leakage through compromised intermediaries. Furthermore, conventional FL frameworks lack intrinsic mechanisms to verify the authenticity and integrity of shared updates, making them inadequate for adversarial and resource-constrained UAV environments. In contrast, the proposed blockchain-integrated distributed ML framework introduces a *trustless, verifiable, and transparent learning environment*. The blockchain layer operates as a decentralized trust infrastructure that records, validates, and synchronizes model parameters through consensus mechanisms and smart contracts. Each UAV node can independently verify contributions from others, ensuring that only authenticated and validated models are incorporated into the global learning process. While this integration introduces computational overhead, it significantly strengthens system resilience, auditability, and operational integrity—qualities indispensable for UAV swarm coordination and autonomous decision-making. The performance evaluation demonstrates that this additional complexity directly enhances system efficacy. Compared to centralized and federated ML models, the proposed architecture achieves:

- **7.6% higher detection accuracy**, reflecting improved reliability in anomaly identification.
- **9.2% lower false positive rate (FPR)**, ensuring more stable decision-making under uncertain network conditions.
- **4.8% higher F1-score**, confirming balanced precision and recall in intrusion detection.

To further clarify the necessity of this complexity, Table 22 provides a comparative analysis of key characteristics across centralized, federated, and blockchain-based ML frameworks.

Overall, the complexity of the proposed system is functionally justified, as it integrates decentralized consensus, verifiable learning, and autonomous coordination—features essential for achieving *secure, scalable, and intelligent UAV operations*. The trade-off between computational cost and operational reliability is balanced through improved fault tolerance, auditability, and predictive accuracy, ensuring that the proposed framework outperforms simpler architectures in both performance and trustworthiness.

5.11. Practical implications for real-world UAV networks

The deployment of the proposed *blockchain-based decentralized ML framework* in real-world UAV networks offers multiple operational advantages and practical insights. The key implications are summarized below:

1. **Decentralized Trust and Data Integrity:** The framework eliminates reliance on a central authority by enabling consensus-based coordination among UAV nodes. This decentralized mechanism ensures that analytical outcomes and model updates are verified and immutable, which is vital in mission-critical domains such as surveillance, reconnaissance, and disaster response.
2. **Collaborative and Privacy-Preserving Learning:** UAV nodes exchange learned model parameters instead of raw sensor data, thereby preserving privacy and reducing bandwidth requirements. This distributed approach allows continuous learning and adaptation to environmental dynamics such as terrain variation, atmospheric changes, and evolving cyber threats.
3. **Enhanced Scalability and Real-Time Responsiveness:** The simulation results demonstrate that the proposed architecture scales linearly with the number of UAV nodes. By storing detailed ML models off-chain and maintaining only essential metadata on-chain, latency is minimized, ensuring real-time decision-making during in-flight operations.
4. **Security and Resilience Against Cyber Threats:** The blockchain backbone safeguards data integrity and prevents post-event manipulation through its immutable ledger. Collaborative anomaly detection enables UAV fleets to identify and isolate compromised nodes quickly, reducing the impact of potential cyberattacks.

5. **Energy and Computational Efficiency:** Although blockchain operations introduce computational overhead, lightweight consensus mechanisms such as *Proof-of-Authority (PoA)* and *Practical Byzantine Fault Tolerance (PBFT)* can be adopted. These approaches maintain trust with reduced energy expenditure, making the framework feasible for resource-constrained UAVs.
6. **Integration with Existing UAV Systems:** The framework can be implemented using embedded controllers such as NVIDIA Jetson or Raspberry Pi with Python- or ROS-based middleware. This modular design allows easy integration with current UAV control systems and mission management architectures.
7. **Operational Robustness in Dynamic Environments:** The system supports adaptive collaboration, enabling UAVs to maintain operational efficiency even under partial communication failure or adversarial network conditions. This resilience makes it suitable for tactical missions and large-scale civilian monitoring.
8. **Improved Intrusion Detection Performance:** By applying stacking-based collaborative reinforcement, UAV nodes achieve higher detection accuracy and lower false positive rates than isolated models. This collective intelligence improves the reliability of intrusion detection in distributed UAV environments.
9. **Implementation Roadmap:** The real-world adoption of this framework can follow a phased deployment strategy: (i) prototype simulation using small UAV clusters, (ii) field testing under limited connectivity scenarios, (iii) integration with air traffic management and defense-grade UAV systems for large-scale validation.

Overall, the proposed framework provides a secure, privacy-preserving, and decentralized solution for UAV-based analytics. Its integration of blockchain and ML enables robust, autonomous, and tamper-resistant aerial intelligence applicable to both civilian and defense sectors.

5.12. Potential limitations and mitigation strategies

While the proposed blockchain-enabled decentralized ML framework demonstrates strong potential for secure and intelligent UAV operations, several practical limitations must be considered. This subsection highlights key challenges and proposes corresponding mitigation strategies to ensure reliable real-world deployment.

1. **Computational Overhead:** Blockchain integration and ML training increase the computational burden on UAV nodes, especially those equipped with low-power processors. *Mitigation:* Lightweight consensus mechanisms such as Proof-of-Authority (PoA) or Practical Byzantine Fault Tolerance (PBFT) can replace resource-intensive Proof-of-Work (PoW). In addition, offloading complex ML tasks to nearby fog or edge servers reduces on-board computation while maintaining decentralized control.
2. **Scalability Challenges:** As the number of UAVs increases, the blockchain ledger size and synchronization overhead may become significant, affecting real-time communication. *Mitigation:* Employing sharding techniques, hierarchical blockchains, or Directed Acyclic Graph (DAG)-based consensus can enhance scalability. Furthermore, implementing sidechains for region-specific UAV clusters minimizes the main chain load while preserving overall system integrity.
3. **Latency in Decision-Making:** The sequential nature of consensus protocols can introduce delays in time-critical UAV missions that require rapid responses. *Mitigation:* Using asynchronous consensus algorithms and off-chain channels for time-sensitive transactions can substantially reduce latency. Additionally, caching frequently accessed data and applying model compression techniques can accelerate local inference.
4. **Energy Consumption:** Continuous participation in blockchain mining and ML model updates increases power usage, reducing UAV flight time. *Mitigation:* Adaptive consensus participation — where UAVs dynamically enter or exit mining roles based on energy status — can balance workload. Integration with renewable or wireless power transfer systems may also sustain longer mission durations.
5. **Data Transmission Bottlenecks:** Exchanging model parameters among multiple UAVs may cause congestion or packet loss, especially in dynamic environments. *Mitigation:* Optimizing communication protocols with edge caching, data compression, and Quality of Service (QoS) control ensures reliable inter-UAV communication without overloading the network.

Overall, by incorporating these mitigation strategies, the proposed framework can achieve a balanced trade-off among computation, scalability, latency, and energy efficiency — thereby ensuring dependable deployment in real-world UAV networks.

6. Conclusion

Industries are increasingly recognizing the value of *Unmanned Aerial Vehicles (UAVs)* or drones, which require the ability to intelligently interpret monitored data for enhanced operational efficiency and real-time decision-making. However, the performance overheads and resource limitations of centralized machine learning (ML) architectures restrict their scalability and efficiency in UAV networks. To address these challenges, this study proposed a **blockchain-enabled decentralized ML framework** that facilitates secure, collaborative, and privacy-preserving analytics among distributed UAV nodes. The novelty of this work lies in integrating blockchain with decentralized ML to enable model-level collaboration without exchanging raw data, thereby ensuring privacy, transparency, and trust in the learning process. The primary contributions of this research include the development of a decentralized architecture that employs blockchain to coordinate secure model sharing, a stacking-based integration mechanism to strengthen predictive performance, and the deployment of smart contracts to automate model validation and consensus among participating nodes. The framework was comprehensively evaluated using the *KDD99 intrusion detection dataset*, where 12,023 records were distributed across blockchain nodes. The results demonstrate that the proposed system achieved an average accuracy of 97.9%, latency 198 ms, and F1-score exceeding 97%, significantly outperforming both centralized and non-stacking models. These outcomes confirm the effectiveness of the proposed framework in enhancing detection accuracy, maintaining data integrity, and supporting scalable distributed analytics in UAV environments. Overall, this study establishes a strong foundation for secure, scalable, and intelligent UAV operations. By sharing trained ML models instead of raw data, the proposed system ensures improved accuracy and reduced privacy risks while maintaining operational efficiency. Furthermore, the blockchain layer adds traceability, immutability, and decentralized control to the collaborative learning process, fostering a trustworthy ecosystem for autonomous aerial analytics.

6.1. Future research directions:

Building upon the promising outcomes of this work, several research avenues can be explored to extend its applicability and robustness:

1. **Scalability and Dataset Diversity:** Future studies can investigate the framework's performance with larger UAV fleets, heterogeneous device configurations, and diverse real-world datasets to evaluate its adaptability and robustness in complex operational environments.

2. **Attack Diversity and Robustness:** Extending the intrusion detection capabilities to handle a broader spectrum of cyber threats — such as zero-day attacks, DDoS, data poisoning, and adversarial ML-based intrusions — would enhance the system's resilience.
3. **Advanced Consensus Mechanisms:** The integration of lightweight and energy-efficient blockchain consensus algorithms such as Proof-of-Stake (PoS), Practical Byzantine Fault Tolerance (PBFT), or Delegated Proof-of-Trust (DPoT) could further reduce latency and computational overhead.
4. **Resource Optimization:** Incorporating adaptive scheduling and model compression techniques may help mitigate computational and communication overheads in resource-constrained UAV networks.
5. **Real-World Deployment:** Future work may also focus on deploying the proposed framework in physical UAV testbeds to analyze performance under dynamic flight conditions, real-time communication delays, and network instability.

CRediT authorship contribution statement

Abdullah Aljumah: Writing – original draft, Supervision, Resources, Methodology, Investigation, Funding acquisition, Data curation. **Tariq Ahamed Ahanger:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Imdad Ullah:** Visualization, Validation, Software, Methodology, Investigation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies

During the preparation of this work, the author(s) used Monica.AI in order to improve English Quality and Grammarly to correct grammatical mistakes. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare no conflict of interest in this area.

Acknowledgments

The authors extend their appreciation to Prince Sattam bin Abdulaziz University for funding this research work through the project number PSAU/2025/01/32567.

Data availability

The data used to support the findings of this study are available from the corresponding author upon request.

References

[1] Mohammad Mozaffari, Walid Saad, Mehdi Bennis, Young-Han Nam, Merouane Debbah, A tutorial on UAVs for wireless networks: Applications, challenges, and open problems, IEEE Commun. Surv. Tutorials 21 (3) (2019) 2334–2360.

[2] Moayad Aloqaily, Rasheed Hussain, Deena Khalaf, Dana Hani, Alma Oracevic, On the role of futuristic technologies in securing UAV-supported autonomous vehicles, IEEE Consum. Electron. Mag. (2022).

[3] Moulay A. Akhloufi, Andy Couturier, Nicolás A. Castro, Unmanned aerial vehicles for wildland fires: Sensing, perception, cooperation and assistance, Drones 5 (1) (2021) 15.

[4] Siti Raihanah Abdani, Mohd Asyraf Zulkifley, Nuraisyah Hani Zulkifley, A lightweight deep learning model for covid-19 detection, in: 2020 IEEE Symposium on Industrial Electronics & Applications, ISIEA, IEEE, 2020, pp. 1–5.

[5] Anuradha Banerjee, Abu Sufian, Krishna Keshob Paul, Sachin Kumar Gupta, EDTP: Energy and delay optimized trajectory planning for UAV-IoT environment, Comput. Netw. 202 (2022).

[6] Petros S Bithas, Emmanouel T Michailidis, Nikolaos Nomikos, Demosthenes Vouyioukas, Athanasios G Kanatas, A survey on machine-learning techniques for UAV-based communications, Sensors 19 (23) (2019) 5170.

[7] Yunfei Chen, Wei Feng, Gan Zheng, Optimum placement of UAV as relays, IEEE Commun. Lett. 22 (2) (2017) 248–251.

[8] Forough Shirin Abkenar, Parisa Ramezani, Saeid Iranmanesh, Sarumathi Murali, Donpiti Chulerttiyawong, Xinyu Wan, Abbas Jamalipour, Raad Raad, A survey on mobility of edge computing networks in IoT: State-of-the-art, architectures, and challenges, IEEE Commun. Surv. Tutorials (2022).

[9] Isis Diaz Linares, Angelife Pardo, Eric Patch, Ali Dehghantanha, Kim-Kwang Raymond Choo, IoT privacy, security and forensics challenges: An unmanned aerial vehicle (UAV) case study, Handb. Big Data Anal. Forensics (2022) 7–39.

[10] Nida Fatima, Paresh Saxena, Manik Gupta, Integration of multi access edge computing with unmanned aerial vehicles: Current techniques, open issues and research directions, Phys. Commun. (2022).

[11] Chunpeng Ge, Willy Susilo, Joonsang Baek, Zhe Liu, Jinyue Xia, Liming Fang, A verifiable and fair attribute-based proxy re-encryption scheme for data sharing in clouds, IEEE Trans. Dependable Secur. Comput. (2021).

[12] Abdullah Ayub Khan, Asif Ali Laghari, Majed Alsafyani, Abdullah M Baqasah, Natalia Kryvinska, Ahmad Almadhor, Roobaea Alroobaea, Michal Gregus Jr., A cost-effective approach using generative AI and gamification to enhance biomedical treatment and real-time biosensor monitoring, Sci. Rep. 15 (1) (2025).

[13] S.M. Asiful Huda, Sangman Moh, Survey on computation offloading in UAV-enabled mobile edge computing, J. Netw. Comput. Appl. (2022).

[14] Ben Nassi, Ron Bitton, Ryusuke Masuoka, Asaf Shabtai, Yuval Elovici, SoK: Security and privacy in the age of commercial drones, in: 2021 IEEE Symposium on Security and Privacy, SP, IEEE, 2021, pp. 1434–1451.

[15] Lav Gupta, Raj Jain, Gabor Vaszkun, Survey of important issues in UAV communication networks, IEEE Commun. Surv. Tutorials 18 (2) (2015) 1123–1152.

[16] Anik Islam, Soo Young Shin, Bus: A blockchain-enabled data acquisition scheme with the assistance of uav swarm in internet of things, IEEE Access 7 (2019) 103231–103249.

[17] Muhammad Firdaus, Kyung-Hyune Rhee, A joint framework to privacy-preserving edge intelligence in vehicular networks, 2022, arXiv preprint arXiv:2208.12755.

[18] Junbo Wang, Amitangshu Pal, Qinglin Yang, Krishna Kant, Kaiming Zhu, Song Guo, Collaborative machine learning: Schemes, robustness, and privacy, IEEE Trans. Neural Netw. Learn. Syst. (2022).

[19] Abdullah Ayub Khan, Asif Ali Laghari, Rex Bacarra, Roobaea Alroobaea, Sultan Algarni, Abdullah M Baqasah, Jamil Abedalrahim Jamil Alsayaydeh, Cybersecurity, digital forensics, and the IoT for deepfake investigation on social media platforms: A review, Human-Centric Comput. Inf. Sci. 15 (2025).

[20] Abdullah Ayub Khan, Asif Ali Laghari, Hamad Almansour, Leila Jamel, Fahima Hajjej, Vania V Estrela, Mohamad Afendee Mohamed, Sajid Ullah, Quantum computing empowering blockchain technology with post quantum resistant cryptography for multimedia data privacy preservation in cloud-enabled public auditing platforms, J. Cloud Comput. 14 (1) (2025) 43.

[21] Abdullah Ayub Khan, Asif Ali Laghari, Roobaea Alroobaea, Abdullah M Baqasah, Majed Alsafyani, Hamed Alsufyani, Sajid Ullah, A lightweight scalable hybrid authentication framework for internet of medical things (IoMT) using blockchain hyperledger consortium network with edge computing, Sci. Rep. 15 (1) (2025).

[22] Abdullah Ayub Khan, Sami Dhabi, Jing Yang, Wajdi Alhakami, Sami Bourouis, Lip Yee, B-LPoET: A middleware lightweight proof-of-elapsed time (PoET) for efficient distributed transaction execution and security on blockchain using multithreading technology, Comput. Electr. Eng. 118 (2024).

[23] Abdullah Ayub Khan, Asif Ali Laghari, Syed Azeem Inam, Sajid Ullah, Muhammad Shahzad, Darakhshan Syed, A survey on multimedia-enabled deepfake detection: state-of-the-art tools and techniques, emerging trends, current challenges & limitations, and future directions, Discov. Comput. 28 (1) (2025) 48.

[24] Asif Ali Laghari, Abdullah Ayub Khan, Amel Ksibi, Fahima Hajjej, Natalia Kryvinska, Ahmad Almadhor, Mohamad Afendee Mohamed, Shtwai Alsubai, A novel and secure artificial intelligence enabled zero trust intrusion detection in industrial internet of things architecture, Sci. Rep. 15 (1) (2025).

[25] Xiaolong Xu, Xuyun Zhang, Honghao Gao, Yuan Xue, Lianyong Qi, Wanchun Dou, Become: Blockchain-enabled computation offloading for IoT in mobile edge computing, IEEE Trans. Ind. Inform. 16 (6) (2019) 4187–4195.

[26] Praveen Pawar, Sanjeev Mani Yadav, Aditya Trivedi, Performance study of dual unmanned aerial vehicles with underlaid device-to-device communications, Wirel. Pers. Commun. 105 (3) (2019) 1111–1132.

[27] Mohammad Saidur Rahman, Ibrahim Khalil, Mohammed Atiquzzaman, Blockchain-powered policy enforcement for ensuring flight compliance in drone-based service systems, IEEE Netw. 35 (1) (2021) 116–123.

[28] Jian Yang, Zhihui Lu, Jie Wu, Smart-toy-edge-computing-oriented data exchange based on blockchain, J. Syst. Archit. 87 (2018) 36–48.

[29] Haya R. Hasan, Khaled Salah, Proof of delivery of digital assets using blockchain and smart contracts, IEEE Access 6 (2018) 65439–65448.

[30] Azade Fotouhi, Haoran Qiang, Ming Ding, Mahbub Hassan, Lorenzo Galati Giordano, Adrian Garcia-Rodriguez, Jinhong Yuan, Survey on UAV cellular communications: Practical aspects, standardization advancements, regulation, and security challenges, IEEE Commun. Surv. Tutorials 21 (4) (2019) 3417–3442.

[31] Israa A. Elnabty, Yasmine Fahmy, Mai Kafafy, A survey on UAV placement optimization for UAV-assisted communication in 5G and beyond networks, Phys. Commun. 51 (2022).

[32] Fan Lai, Yinwei Dai, Sanjay Singapuram, Jiachen Liu, Xiangfeng Zhu, Harsha Madhyastha, Mosharaf Chowdhury, Fedscale: Benchmarking model and system performance of federated learning at scale, in: International Conference on Machine Learning, PMLR, 2022, pp. 11814–11827.

[33] Yulan Gao, Ziqiang Ye, Han Yu, Zehui Xiong, Yue Xiao, Dusit Niyato, Multi-resource allocation for on-device distributed federated learning systems, 2022, arXiv preprint arXiv:2211.00481.

[34] Meiguang Zheng, Dongbang Mao, Liu Yang, Yeming Wei, Zhigang Hu, DOSP: an optimal synchronization of parameter server for distributed machine learning, J. Supercomput. (2022) 1–28.

[35] Sheng Shen, Tianqing Zhu, Di Wu, Wei Wang, Wanlei Zhou, From distributed machine learning to federated learning: In the view of data privacy and security, Concurr. Comput.: Pr. Exp. 34 (16) (2022) e6002.

[36] Aditya Ajgaonkar, Anuj Raghani, Bhavya Sheth, Dyuwan Shukla, Dhiren Patel, Sanket Shanbhag, A blockchain approach for exchanging machine learning solutions over smart contracts, in: Science and Information Conference, Springer, 2022, pp. 470–482.

[37] Narander Kumar, Jitendra Kumar Samriya, Blockchain-based efficient resource allocation and minimum energy utilization in cloud computing using SMO, in: Blockchain for 6G-Enabled Network-Based Applications, CRC Press, pp. 129–157.

[38] Peng Zhai, Jingsha He, Nafei Zhu, Blockchain-based internet of things access control technology in intelligent manufacturing, Appl. Sci. 12 (7) (2022) 3692.

[39] Hamza Baniata, Radu Prodan, Attila Kertesz, Machine learning for alternative mining in pow-based blockchains: Theory, implications and applications, 2022.

[40] Shixiang Lu, Zhiwei Gao, Qifa Xu, Cuixia Jiang, Aihua Zhang, Xiangxiang Wang, Class-imbalance privacy-preserving federated learning for decentralized fault diagnosis with biometric authentication, IEEE Trans. Ind. Inform. 18 (12) (2022) 9101–9111.

[41] Zakria Qadir, Fahim Ullah, Hafiz Suliman Munawar, Fadi Al-Turjman, Addressing disasters in smart cities through UAVs path planning and 5G communications: A systematic review, Comput. Commun. 168 (2021) 114–135.

[42] Hamman A. Bimmo, Budi Rahardjo, Fed-CALiBER: Federated lightweight BERT intrusion detection on CAN bus protocol in autonomous vehicle, IEEE Access (2025).

[43] Ji Sun Shin, et al., SAFE-IDS: A privacy-preserving framework for overcoming non-IID challenges in federated intrusion detection, Comput. Secur. (2025) 104492.

[44] Abdelaziz Mohamed Ali Salama, Decentralised Federated Learning over Wireless Communication Networks (Ph.D. thesis), University of Leeds, 2024.